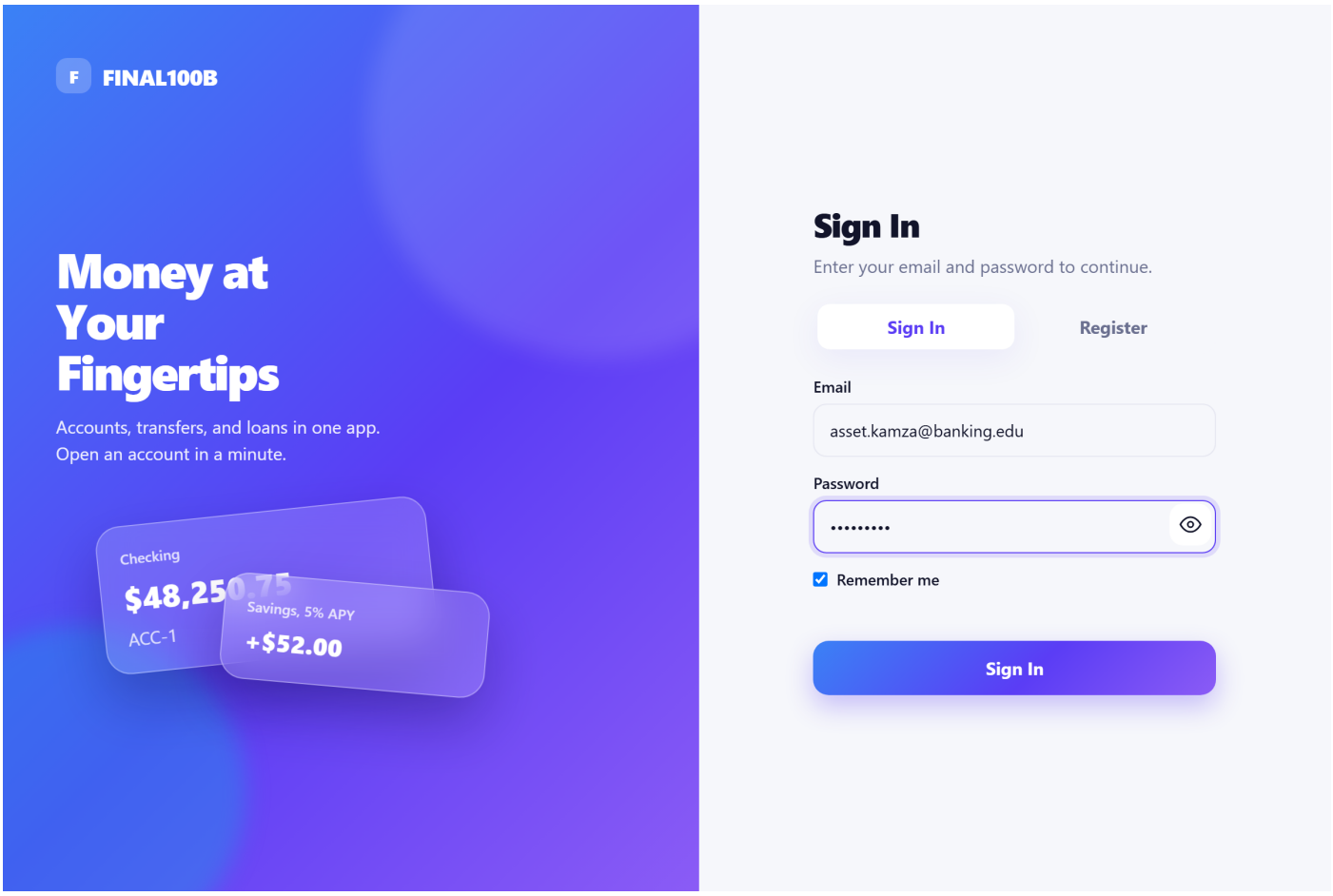
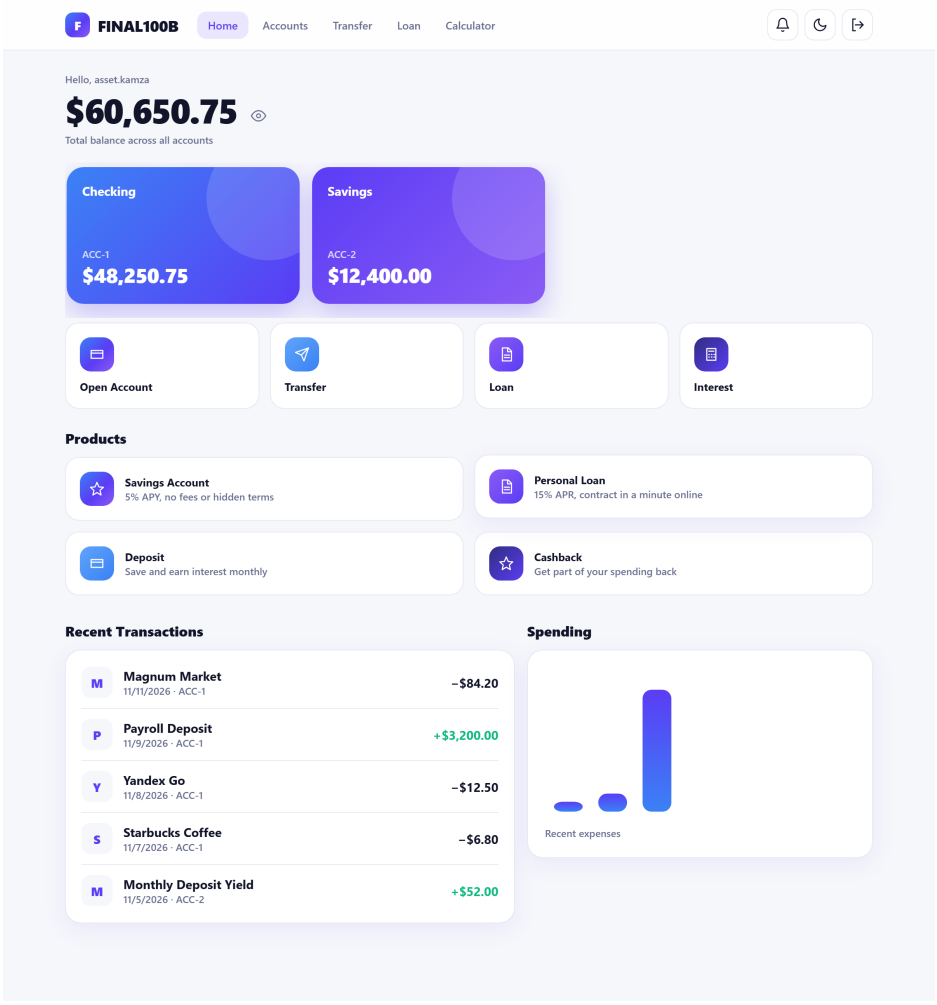
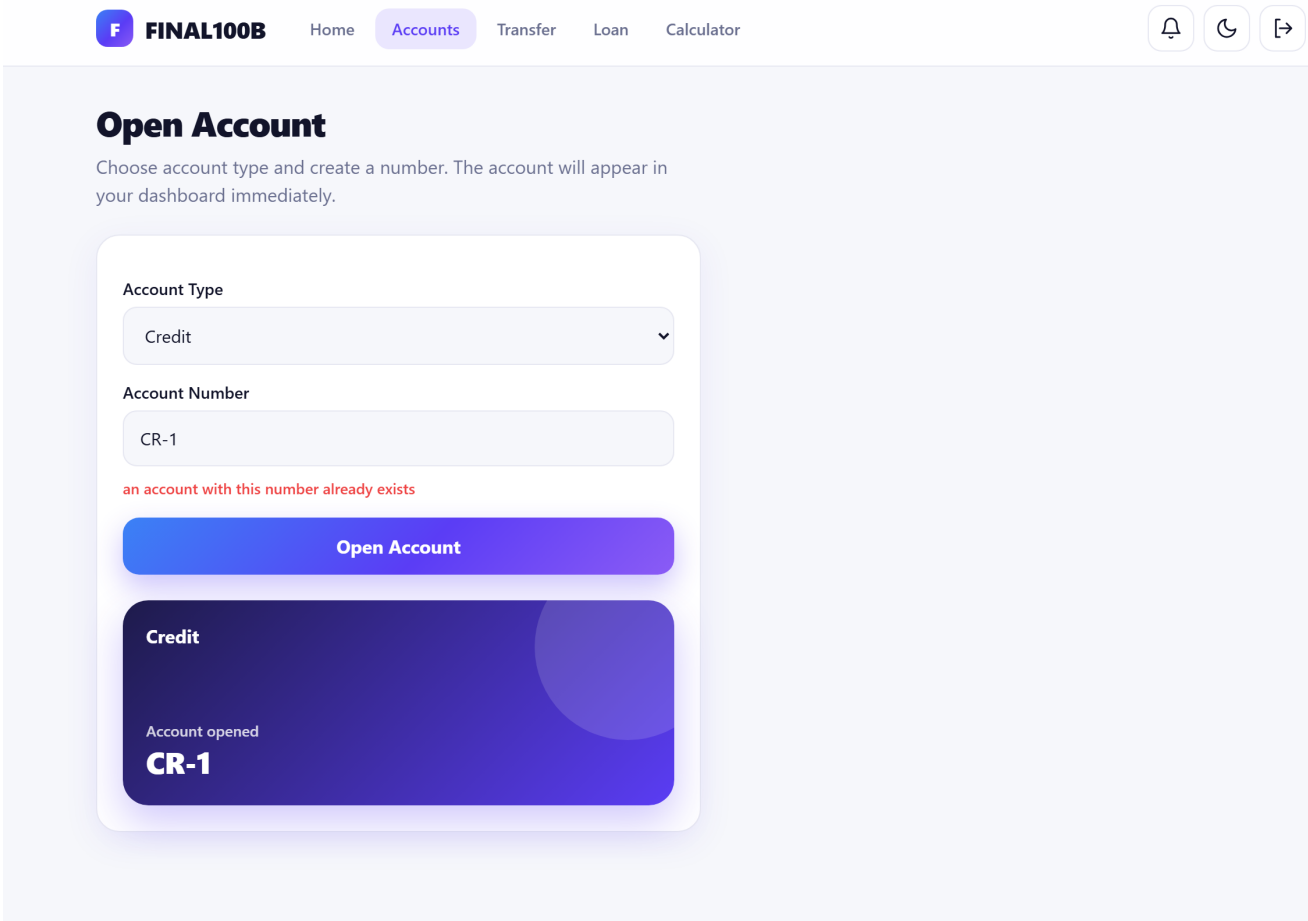
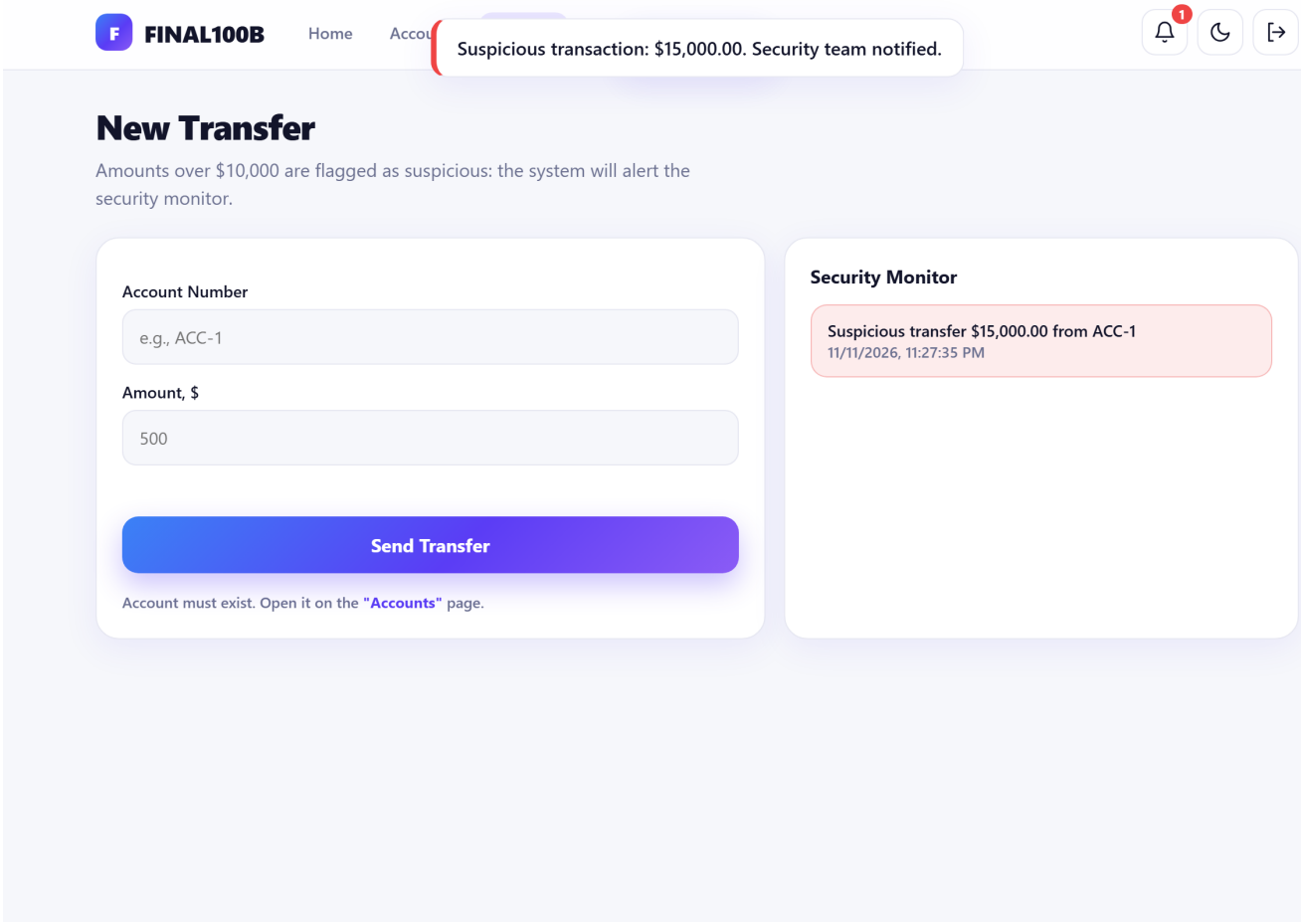
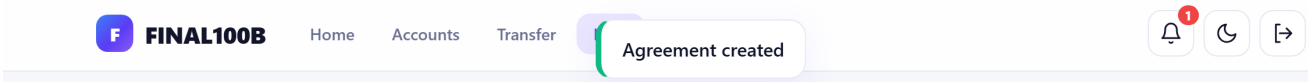

FINAL100B
Money at Your Fingertips
Accounts, transfers, and loans in one app.
Open an account in a minute.
Checking
$48,250.75
ACC-1
Savings, 5% APY
+$52.00
Sign In
Enter your email and password to continue.
Sign In
Register
Email
asset.kamza@banking.edu
Password
Remember me
Sign In


FINAL100B
Home
Accounts
Transfer
Loan
Calculator
Hello, asset.kamza
$60,650.75
Total balance across all accounts
Checking
ACC-1
$48,250.75
Savings
ACC-2
$12,400.00
Open Account
Transfer
Loan
Interest
Products
Savings Account
5% APY, no fees or hidden terms
Personal Loan
15% APR, contract in a minute online
Deposit
Save and earn interest monthly
Cashback
Get part of your spending back
Recent Transactions
Magnum Market
11/11/2026 · ACC-1
−$84.20
Payroll Deposit
11/9/2026 · ACC-1
+$3,200.00
Yandex Go
11/8/2026 · ACC-1
−$12.50
Starbucks Coffee
11/7/2026 · ACC-1
−$6.80
Monthly Deposit Yield
11/5/2026 · ACC-2
+$52.00
Spending
Recent expenses


FINAL100B
Home
Accounts
Transfer
Loan
Calculator
Open Account
Choose account type and create a number. The account will appear in your dashboard immediately.
Account Type
Credit
Account Number
CR-1
an account with this number already exists
Open Account
Credit
Account opened
CR-1


FINAL100B
Home
Suspicious transaction: $15,000.00. Security team notified.
1
New Transfer
Amounts over $10,000 are flagged as suspicious: the system will alert the security monitor.
Account Number
e.g., ACC-1
Amount, $
500
Send Transfer
Account must exist. Open it on the "Accounts" page.
Security Monitor
Suspicious transfer $15,000.00 from ACC-1
11/11/2026, 11:27:35 PM



Apply for Loan

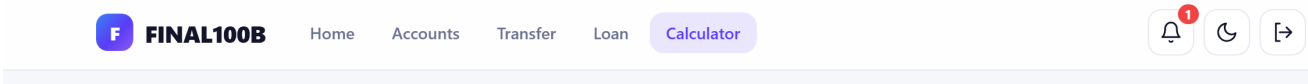
Contract is built step by step: borrower, amount, standard terms.

New Application

Loan Agreement LN-683004

Date: 11/11/2026
Borrower: Asset Kamza
Amount: $5,000.00
Rate: 15% APR, term 12 months
Monthly payment: **$451.29**
Total to repay: **$5,415.50**

Print Agreement



Interest Calculator

Product type selects the calculation strategy: savings (5%) or credit (15%).

Type

Savings

Amount, $

1000

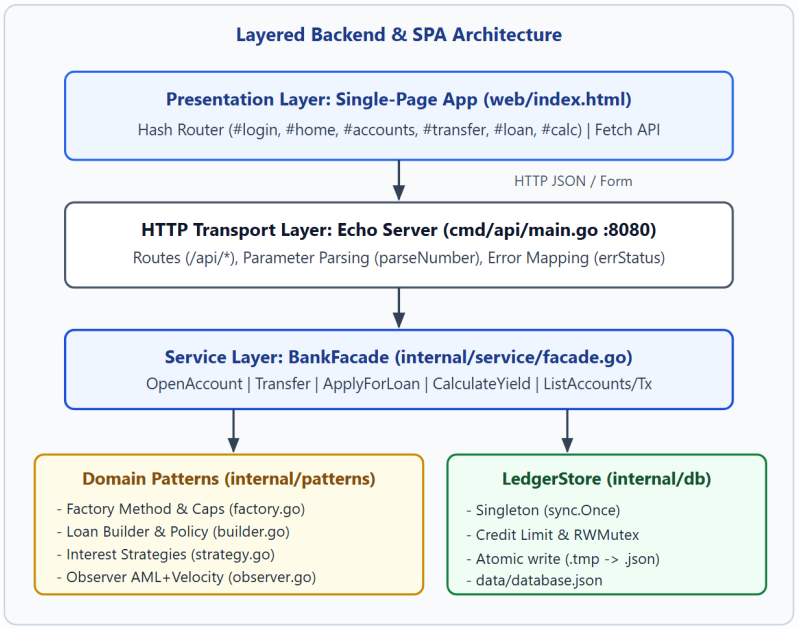
Calculate

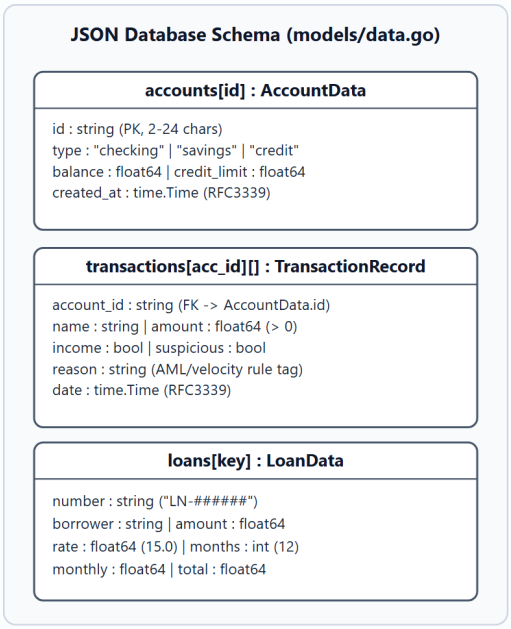
Annual interest (5%)

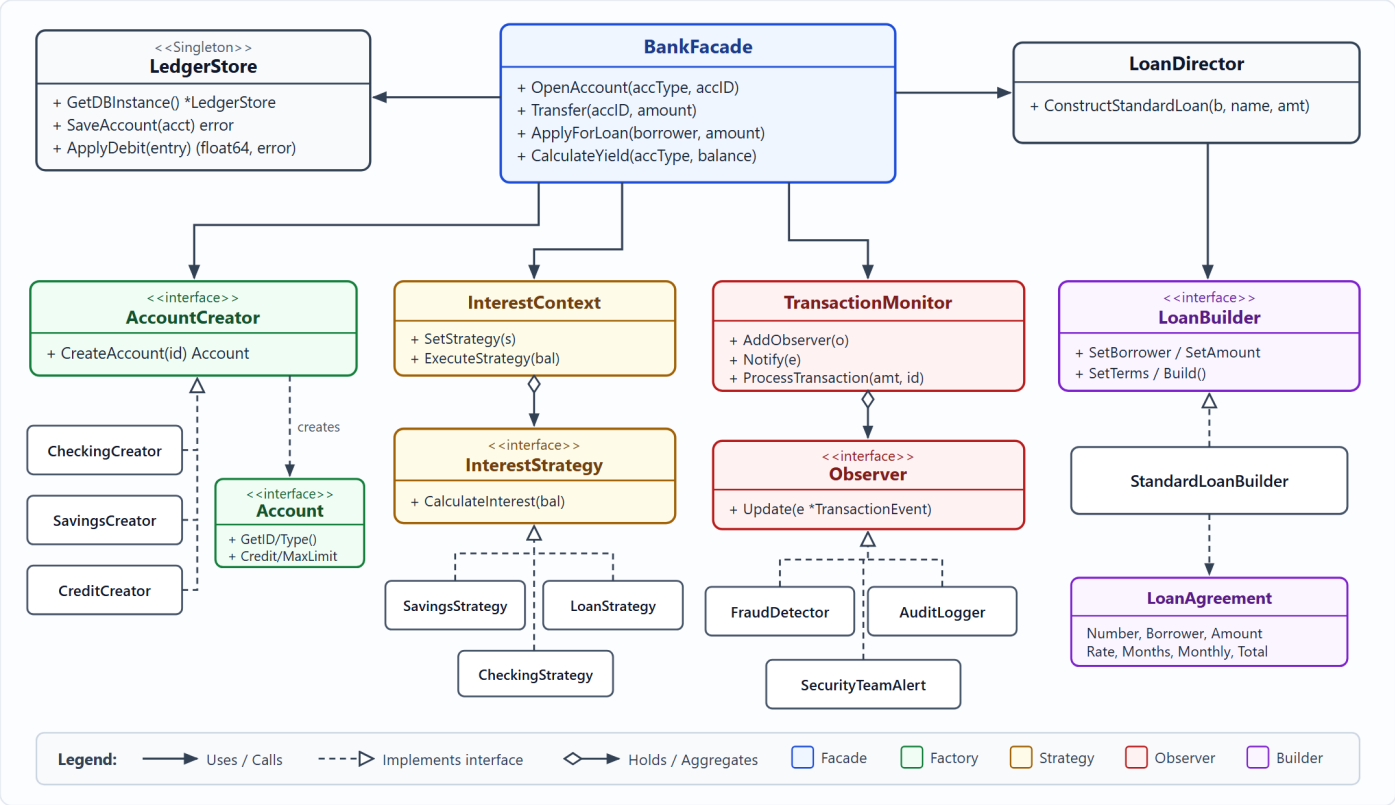
$50.00

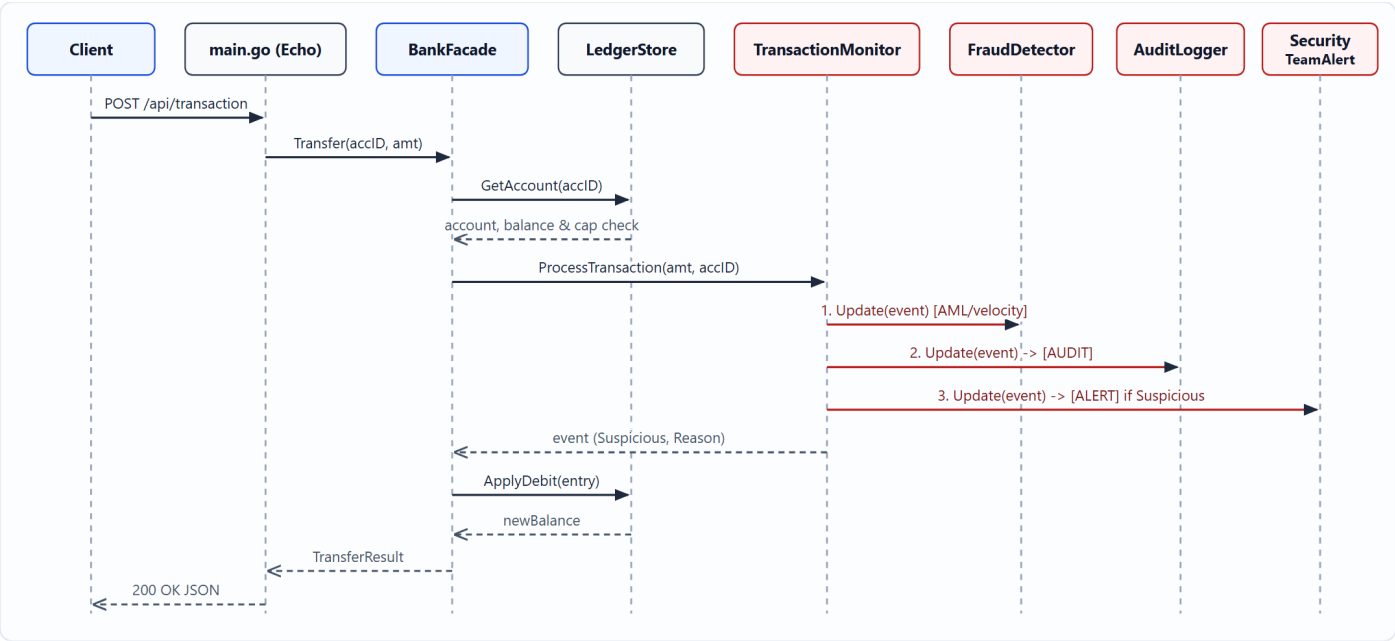
Total after one year: **$1,050.00**

System Architecture & Data Schema Overview






<<Singleton>>
LedgerStore
+ GetDBInstance() *LedgerStore
+ SaveAccount(acct) error
+ ApplyDebit(entry) (float64, error)
BankFacade
+ OpenAccount(accType, accID)
+ Transfer(accID, amount)
+ ApplyForLoan(borrower, amount)
+ CalculateYield(accType, balance)
LoanDirector
+ ConstructStandardLoan(b, name, amt)
<<interface>>
AccountCreator
+ CreateAccount(id) Account
creates
CheckingCreator
SavingsCreator
CreditCreator
<<interface>>
Account
+ GetID/Type()
+ Credit/MaxLimit
InterestContext
+ SetStrategy(s)
+ ExecuteStrategy(bal)
<<interface>>
InterestStrategy
+ CalculateInterest(bal)
SavingsStrategy
LoanStrategy
CheckingStrategy
TransactionMonitor
+ AddObserver(o)
+ Notify(e)
+ ProcessTransaction(amt, id)
<<interface>>
Observer
+ Update(e *TransactionEvent)
FraudDetector
AuditLogger
SecurityTeamAlert
<<interface>>
LoanBuilder
+ SetBorrower / SetAmount
+ SetTerms / Build()
StandardLoanBuilder
LoanAgreement
Number, Borrower, Amount
Rate, Months, Monthly, Total
Legend:
Uses / Calls
Implements interface
Holds / Aggregates
Facade
Factory
Strategy
Observer
Builder


Client
main.go (Echo)
BankFacade
LedgerStore
TransactionMonitor
FraudDetector
AuditLogger
Security
TeamAlert
POST /api/transaction
Transfer(accID, amt)
GetAccount(accID)
account, balance & cap check
ProcessTransaction(amt, accID)
1. Update(event) [AML/velocity]
2. Update(event) -> [AUDIT]
3. Update(event) -> [ALERT] if Suspicious
event (Suspicious, Reason)
ApplyDebit(entry)
newBalance
TransferResult
200 OK JSON